

d1: Scaling Reasoning in Diffusion Large Language Models via Reinforcement Learning

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Abstract

Recent large language models (LLMs) have demonstrated strong reasoning capabilities that benefit from online reinforcement learning (RL). These capabilities have primarily been demonstrated within the left-to-right autoregressive (AR) generation paradigm. In contrast, non-autoregressive paradigms based on diffusion generate text in a coarse-to-fine manner. Although recent diffusion-based large language models (dLLMs) have achieved competitive language modeling performance compared to their AR counterparts, it remains unclear if dLLMs can also leverage recent advances in LLM reasoning. To this end, we propose *d1*, a framework to adapt pre-trained masked dLLMs into reasoning models via a combination of supervised finetuning (SFT) and RL. Specifically, we develop and extend techniques to improve reasoning in pretrained dLLMs: (a) we utilize a masked SFT technique to distill knowledge and instill self-improvement behavior directly from existing datasets, and (b) we introduce a novel critic-free, policy-gradient based RL algorithm called *diffu*-GRPO, the first integration of policy gradient methods to masked dLLMs. Through empirical studies, we investigate the performance of different post-training recipes on multiple mathematical and planning benchmarks. We find that *d1* yields the best performance and significantly improves performance of a state-of-the-art dLLM. Our code is released at <https://d1lm-reasoning.github.io/>.

1 Introduction

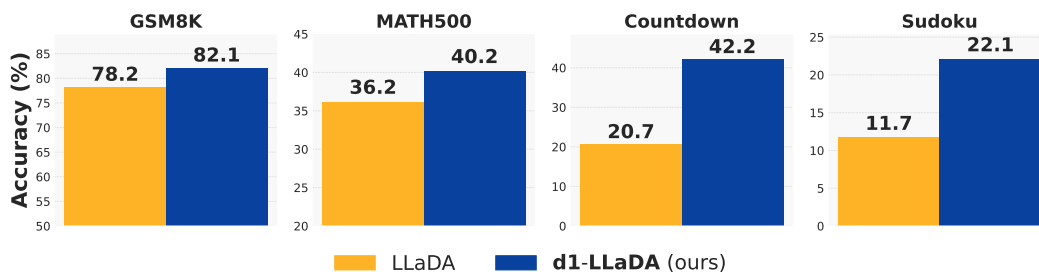


Figure 1: Across four math and planning tasks, **d1-LLaDA**, which undergoes SFT followed by our proposed *diffu*-GRPO, consistently outperforms the base LLaDA-8B-Instruct model. We report results using the best performing generation sequence length for each task and model, with complete sequence length results shown in Table 1.

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Recent advances in large language models (LLMs) have demonstrated remarkable capabilities across diverse applications spanning chatbots, coding, summarization, and translation [1, 13]. While these models typically scale through next-token prediction on vast corpora via computationally intensive pretraining, the finite availability of high-quality training data poses a fundamental scaling challenge. Reinforcement learning (RL) methods have emerged as a promising post-training method, enabling models to generate and explore with reward signals rather than relying solely on static datasets. This approach has yielded significant improvements on reasoning tasks in recent models, such as DeepSeek-R1 [17] and Kimi K1.5 [41], demonstrating that applying RL directly to base models can achieve performance comparable to OpenAI’s o1 model [31]. However, these advances in RL-based post-training have primarily been limited to autoregressive LLMs that operate through left-to-right, sequential inference.

In a parallel line of work, discrete diffusion large language models (dLLMs) [30, 15, 29, 48] have emerged as promising non-autoregressive alternatives for language modeling. Unlike AR models that generate text token-by-token in a causal manner, dLLMs generate text through an iterative denoising process, refining sequences over multiple steps while leveraging both past and future context via bidirectional attention. Among them, open masked dLLMs such as LLaDA [30] have demonstrated performance comparable to similarly sized AR models, and closed-source dLLMs such as Mercury [20] further demonstrate excellent inference efficiency. However, leading open-source dLLMs have not undergone RL post-training, leaving this promising direction largely unexplored. This paradigm shift raises important questions about how RL post-training might be effectively realized in a non-autoregressive context.

Adapting RL algorithms to masked dLLMs poses unique challenges because existing successful approaches for AR models, such as PPO [37] and GRPO [38], rely on estimating and optimizing policy distributions through computing log-probabilities of generated sequences, which cannot be directly applied to dLLMs. While this computation is straightforward in AR models through sequential factorization, dLLMs lack this natural decomposition due to their iterative, non-sequential generation process.

To bridge this gap, **we propose d1, a two-stage post-training framework for enhancing reasoning in masked dLLMs.** In the first stage, the model undergoes supervised finetuning (SFT) on high-quality reasoning traces. **In the RL stage, we introduce *diffu*-GRPO**, a novel policy gradient method for masked dLLMs that builds upon GRPO with our proposed efficient one-step estimation of log-probabilities. To the best of our knowledge, this represents the first application of policy gradient RL to masked dLLMs. Our estimator leverages random prompt masking, which acts a form of regularization for policy optimization, allowing us to scale the number of gradient updates per batch and reduces the number of online generations required by RL training. This substantially reduces the compute time.

Empirically, we instantiate d1 using LLaDA-8B-Instruct as our base model. We compare d1-LLaDA’s performance with the base LLaDA model, as well as with LLaDA variants trained using SFT-only and *diffu*-GRPO-only approaches. Our experiments demonstrate that d1 consistently outperforms the base model across four reasoning tasks in math and planning, as shown in Figure 1, with nearly doubled performance on planning tasks. Furthermore, d1 surpasses both the SFT-only and *diffu*-GRPO-only methods. Additionally, we complement our primary findings with thorough ablation studies on algorithm design, qualitative analysis, and extensions of *diffu*-GRPO to coding tasks, where we also observe consistent improvements.

2 Preliminaries

2.1 Masked Diffusion Large Language Models

Masked dLLMs [5, 36, 39, 32, 26], involve a forward process that gradually corrupts a sequence of tokens x_0 by the mask token. The process is indexed by time $t \in [0, 1]$. At timestep t , the sequence x_t is partially masked, where for each token the probability of remaining unmasked is α_t . Particularly, α_t (a.k.a noise schedule) is strictly decreasing in t . When $t = 1$, all the tokens in x_1 are masked. To train a masked dLLM, we begin by designing a forward process with a specific form of α_t . We parameterize a bidirectional unmasking predictor f_θ . In each iteration, we randomly sample a timestep $t \in [0, 1)$ and mask the tokens based on the designed forward process. Given these

corrupted inputs, the learning objective is to predict the original tokens. The standard loss function for this task is the negative evidence lower bound (NELBO), which is an upper bound of the negative log-likelihood (NLL) of the data. For masked dLLMs, NELBO simplifies to a weighted NLL, where the weights are determined by a transformation of α_t [36, Equation (10)]. In this work, we apply d1 on top of LLaDA [30], whose forward process sets $\alpha_t = 1 - t$ and the resulting NELBO is

$$-\mathbb{E}_{t \sim \mathcal{U}[0,1], x_0 \sim p_{\text{data}}, x_t \sim q_{t|0}(x_t|x_0)} \left[\frac{1}{t} \sum_{k=1}^{|x_t|} \mathbb{1}[x_t^k = \text{mask}] \log f_{\theta}(x_0^k | x_t) \right], \quad (1)$$

where $|x_t|$ is the sequence length of x , and x^k is the k -th token. Note that the loss is only calculated for tokens that are masked out in timestep t . The key difference between masked dLLMs and BERT [12] is that the latter uses a fixed masking ratio and the decoding is a single-step infilling process, whereas masked dLLMs use time-varying masking ratios and the decoding process involves multiple steps starting from pure noise and thus resulting in a generative model. Further details about the formulation of masked dLLMs are deferred to Appendix C.

2.2 Group Relative Policy Optimization for Large Language Models

Policy gradient methods have been widely adopted in the post-training stage to enhance the performance of LLMs [33, 7, 22, 2]. While Proximal Policy Optimization (PPO) [37] has been the predominant approach in online RL, it requires jointly training a state value function V to estimate advantages, leading to increased computational demands. Group Relative Policy Optimization (GRPO) [38] offers a more efficient alternative by using group statistics to derive advantages. For each question q , GRPO samples a group of G responses $\{o_1, o_2, \dots, o_G\}$ from the old policy $\pi_{\theta_{\text{old}}}$. It then sets the advantages for all tokens $k = 1, \dots, |o_i|$ for o_i as the normalized reward $\frac{r_i - \text{mean}(\{r_j\}_{j=1}^G)}{\text{std}(\{r_j\}_{j=1}^G)}$.

Here, we can view $\text{mean}(\{r_j\}_{j=1}^G)$ as a G -sample Monte Carlo estimation of the value $V(q)$, while the sparse reward r_i serves as the (undiscounted) state-action value $Q(q, o_i)$. However, normalizing the advantage $Q(q, o_i) - V(q)$ by nonzero state function introduces bias into policy gradient estimation. Therefore, similar to Liu et al. [24], we use the unnormalized advantage

$$A_i^k(\pi) = r_i(\pi) - \text{mean}(\{r_j(\pi)\}_{j=1}^G), \quad 1 \leq k \leq |o_i|. \quad (2)$$

The rest of our RL setup follows GRPO. The objective function incorporates a clipping mechanism (similar to PPO) to moderate policy updates, and a reverse KL penalty to prevent excessive deviation from the reference policy:

$$\mathcal{L}_{\text{GRPO}}(\theta) = \mathbb{E}_{\substack{q \sim \mathcal{D} \\ o_1, \dots, o_G \sim \pi_{\theta}(\cdot|q)}} \left[\left(\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{k=1}^{|o_i|} \min(\rho_i^k A_i^k, \text{clip}(\rho_i^k, 1 - \varepsilon, 1 + \varepsilon) A_i^k) \right) - \beta D_{\text{KL}}[\pi_{\theta}(\cdot|q) \parallel \pi_{\text{ref}}(\cdot|q)] \right], \quad (3)$$

where π_{θ} is the current policy being updated, $\pi_{\theta_{\text{old}}}$ is the policy before the update, $\rho_i^k = \frac{\pi_{\theta}(o_i^k|q, o_i^{<k})}{\pi_{\theta_{\text{old}}}(o_i^k|q, o_i^{<k})}$, A_i^k is computed using $\pi_{\theta_{\text{old}}}$ and Equation (2), and π_{ref} is the reference policy (typically the initial model). The clipping parameter ε limits the magnitude of policy updates to ensure stability, while β controls the strength of the KL divergence regularization.

3 d1: Adapting Pre-trained Masked dLLMs to Reasoning Models

We propose d1, a two-stage framework that enhances the reasoning performance of pre-trained masked dLLMs by sequentially combining SFT and online RL.

Online RL, particularly the GRPO algorithm, has demonstrated its efficacy in improving the performance of offline trained language model [38, 17, 41]. However, the learning formulation of GRPO does not directly generalize to dLLMs. The objective of GRPO (3) requires computing the (log-)likelihood ratio of π_{θ} and $\pi_{\theta_{\text{old}}}$, at both the token level (for the advantage weights) and the sequence level (for the reverse KL term). Generally speaking, we need to efficiently compute the per-token and the sequence log-probability of dLLMs' completion o . Autoregressive (AR) models, such as Transformers, directly model the per-token log-probabilities, and the sequence-level log-probability of o can be easily computed through the chain rule using one forward pass: $\log \pi_{\text{AR}}(o|q) = \sum_{k=1}^{|o|} \log \pi_{\text{AR}}(o^k|q, o^{<k})$. Similarly, the KL term can be decomposed